

DRY-RUN GUIDE | two_pointers_arr.py
Source: Python\3rd_Belt\two_pointers_arr.py

CORE BEHAVIOR

Prints the distinct union of two sorted arrays using advancing pointers and skipping duplicate runs.

REPRESENTATIVE INPUT

'1 2 2 4' and '2 3 4'.

STEP TRACE

Emit 1; match 2 and emit once while skipping its duplicate; emit 3; emit 4 once.

EXPECTED OUTPUT / RETURN

'1 2 3 4'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\two_pointers_arr.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Use the two-pointer technique to solve the array problem.
```

```
002 | # Add your solution here.
```

```
003 |
```

```
004 | def find_union_sorted(arr1, arr2):
```

```
005 |     i, j = 0, 0
```

```
006 |     first = True
```

```
007 |
```

```
008 |     while i < len(arr1) and j < len(arr2):
```

```
009 |
```

```
010 |         if arr1[i] < arr2[j]:
```

```
011 |             value = arr1[i]
```

```
012 |
```

```
013 |             if first:
```

```
014 |                 print(value, end="")
```

```
015 |                 first = False
```

```
016 |             else:
```

```
017 |                 print(" " + str(value), end="")
```

```
018 |
```

```
019 |             while i < len(arr1) and arr1[i] == value:
```

```
020 |                 i += 1
```

```
021 |
```

```
022 |             elif arr2[j] < arr1[i]:
```

```
023 |                 value = arr2[j]
```

```
024 |
```

```
025 |             if first:
```

```
026 |                 print(value, end="")
```

```
027 |                 first = False
```

```
028 |             else:
```

```
029 |                 print(" " + str(value), end="")
```

```
030 |
```

```
031 |             while j < len(arr2) and arr2[j] == value:
```

```
032 |                 j += 1
```

```
033 |
```

```
034 |             else:
```

```
035 |                 value = arr1[i]
```

```
036 |
```

```
037 |         if first:
038 |             print(value, end="")
039 |             first = False
040 |         else:
041 |             print(" " + str(value), end="")
042 |
... 31 later source lines omitted; inspect the original.
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.